6. Interdisplinary Modeling in Quantitative Trading (Part 1/3)

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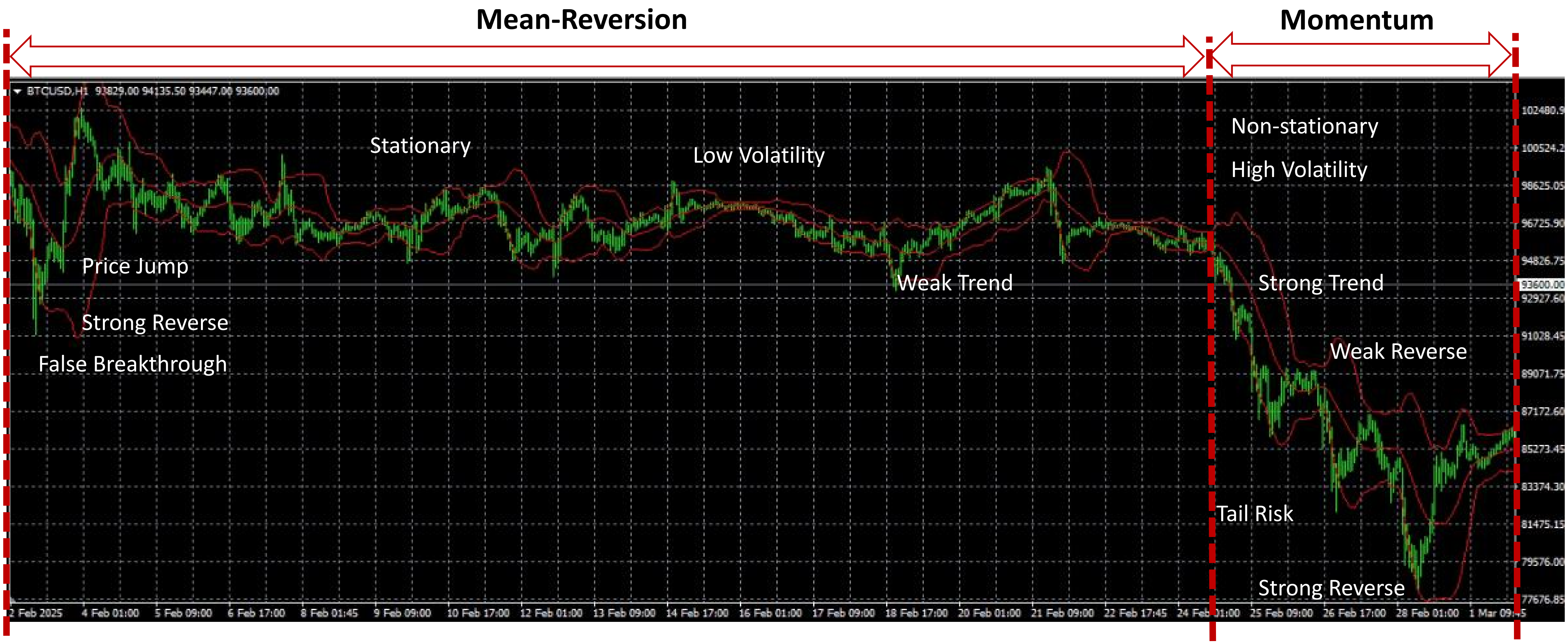
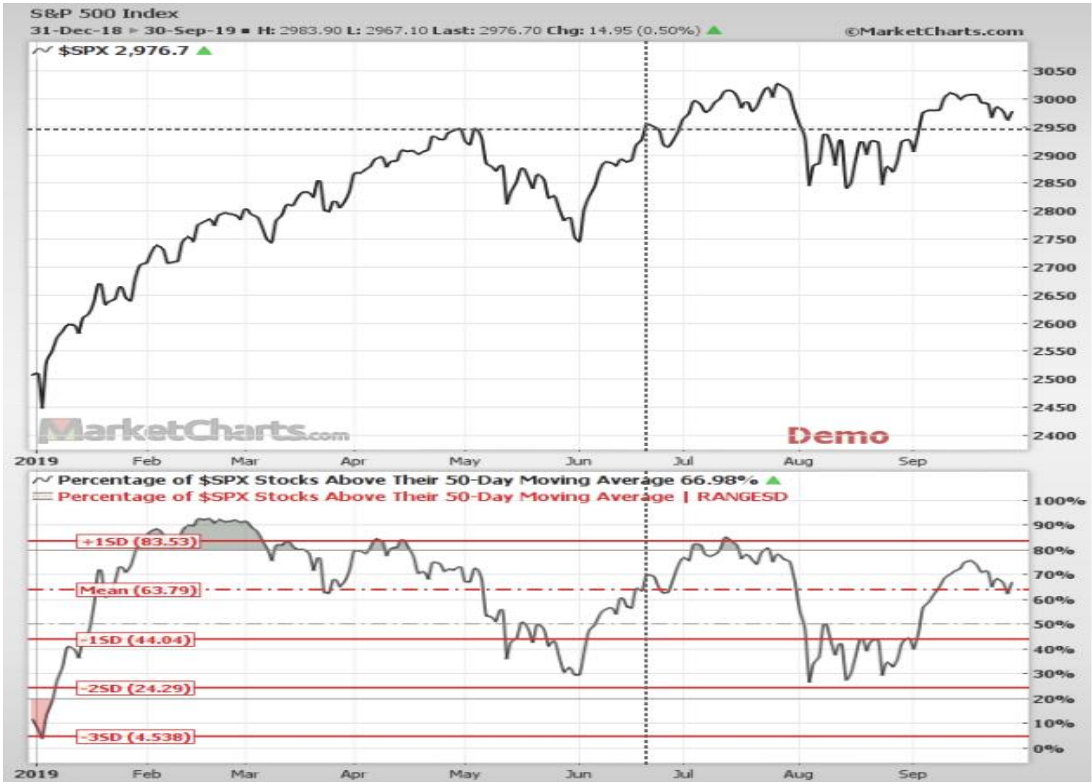
CONTENTS

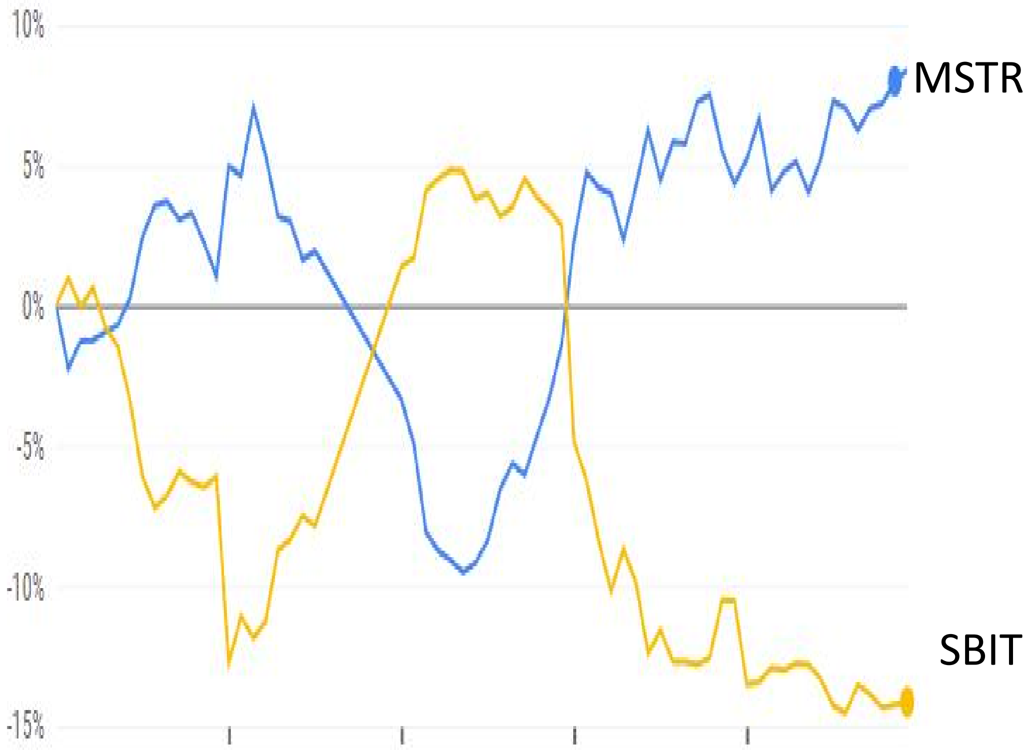
1. Core Challenges in Modeling a Quantitative Strategy: Momentum vs Mean-reversion



Type	Profitability	Winning Probabilibty	Trading Frequency	Tail Risk
Momentum	High	Low	Low	Low
Mean-Reversion	Low	High	High	High

1. Core Challenges in Modeling a Quantitative Strategy: Co-movement





Breadth Indicator (Left): percentage of stocks above certain price level.
Intraday co-movement of MSTR and SBIT (Right): When the underlying price is stationary, pairs trading between leveraged products is profitable.

1. Core Challenges in Modeling a Quantitative Strategy: Predictability of the Market

- **Market predictability is a necessary empirical condition for a quantitative model to be considered reliable and robust.**
- The predictability measure operates as a regime selector, distinguishing between momentum and mean-reversion intervals.
- Predictability is related to the concepts:

Stationarity, Randomness, Degree of Chaos, Measure of Complexity, Measure of Market Efficiency, etc.

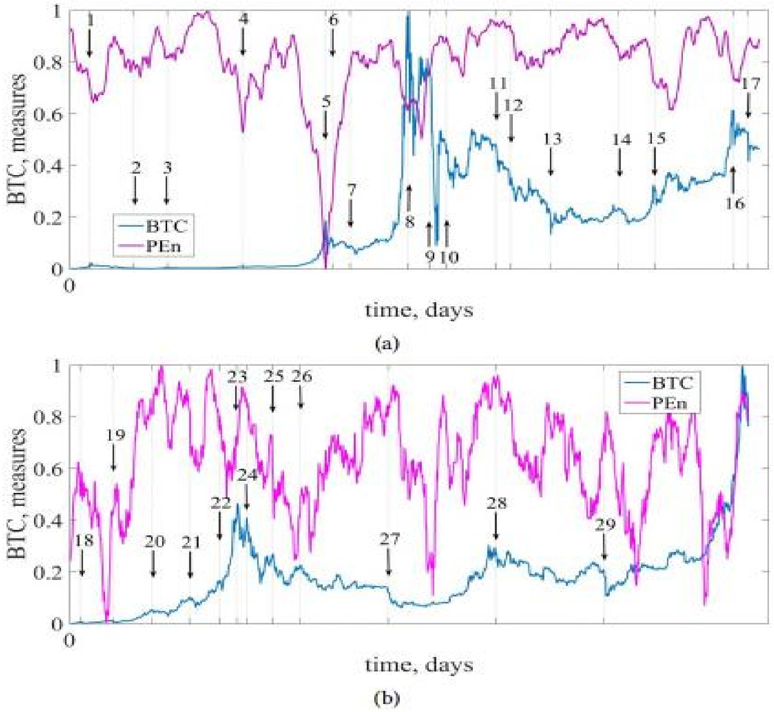


Figure 9: PEn dynamics along with the first (a) and the second (b) periods of the entire time series of Bitcoin.

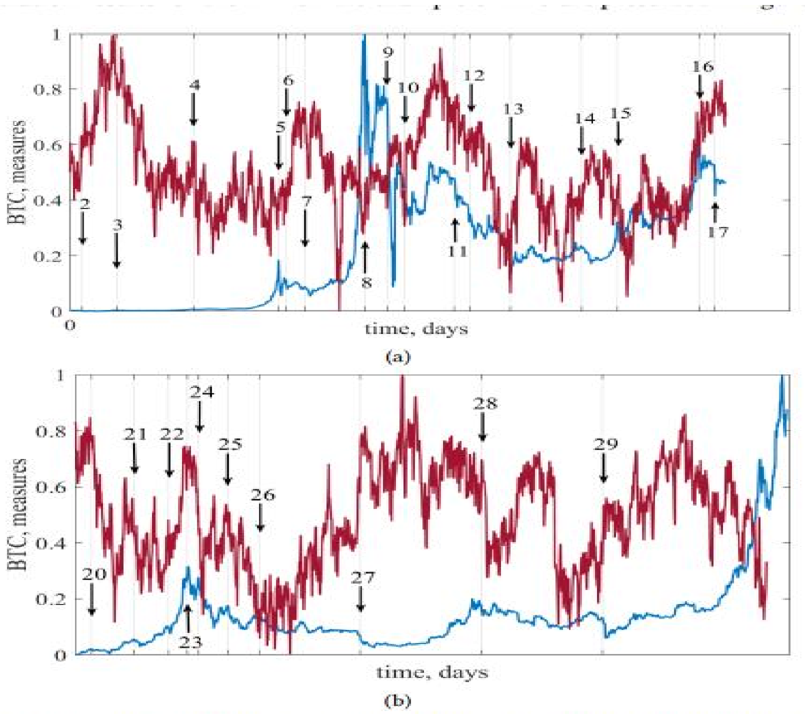


Figure 14: The dynamics of the LLE (λ_{max}) with Eckmann et al. method within the time window of the length 250 days and the time step of 1 day for the first (a) and second (b) periods. Exponents are calculated with $d_E = 3$ and $d_M = 2$.

Source: Bielinskyi, A. O., & Serdyuk, O. A. (2021). Econophysics of cryptocurrency crashes: a systematic review.

2. Interdisplinary Modeling: Momentum (Examples in Technical Analysis)

- MA, MACD are directional trend indicators.
- ADX is non-directional. It only measures the strength of the trend.
- The performance is sensitive to the parameter selected (Time window, threshold)
- Fake cross-overs generate losing trades.
- Selecting a time window involves a direct trade-off: longer windows filter noise more effectively but introduce lag.

2. Interdisplinary Modeling: Momentum (Example in Statistics)

Let's assume:

- a price series $\{p_t\} = \{p_0, p_1, p_2, \ldots, p_T\}$ and
- a log return series $\{x_t\} = \{x_1, x_2, \ldots, x_T\}$ where $x_t = \ln \frac{p_t}{p_{t-1}}$

Variance Ratio (VR)

The **variance ratio** of k-period return is defined as:

$$V(k) = \frac{Var(x_t + x_{t-1} + \ldots + x_{t-k+1})/k}{Var(x_t)}$$

The estimator of $V(k)$ proposed in Lo and MacKinlay (1988) is

$$VR(k) = \frac{\hat{\sigma}^2(k)}{\hat{\sigma}^2(1)}$$

where $\hat{\sigma}^2(1)$ is the unbiased estimator of the one-period return variance, using the one-period returns $\{x_t\}$, and is defined as

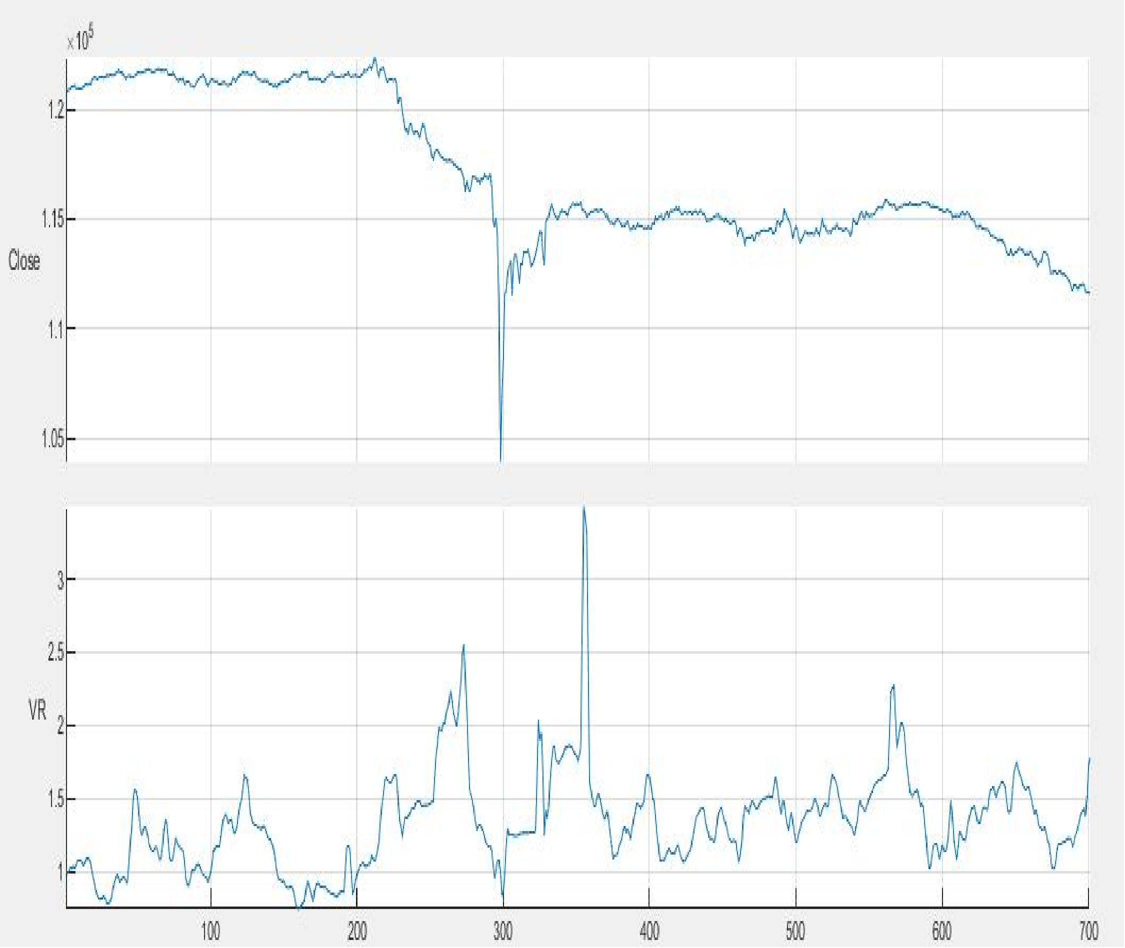
$$\hat{\sigma}^2(1) = \frac{1}{T-1} \sum_{t-1}^{T} (x_t - \hat{\mu})^2$$

and $\hat{\sigma}^2(k)$ is the estimator of k-period return variance using k-period returns. Lo and MacKinlay (1988) defined it, due to limited sample size and the desire to improve the power of the test, as

$$\hat{\sigma}^2(k) = \frac{1}{m} \sum_{t-1}^{T} \left(\ln \frac{P_t}{P_{t-k}} - k\hat{\mu} \right)^2$$

where $m = k(T-k+1)(1-k/T)$ is chosen such that $\hat{\sigma}^2(k)$ is an unbiased estimator of the k-period return variance when σ_t^2 is constant over time.

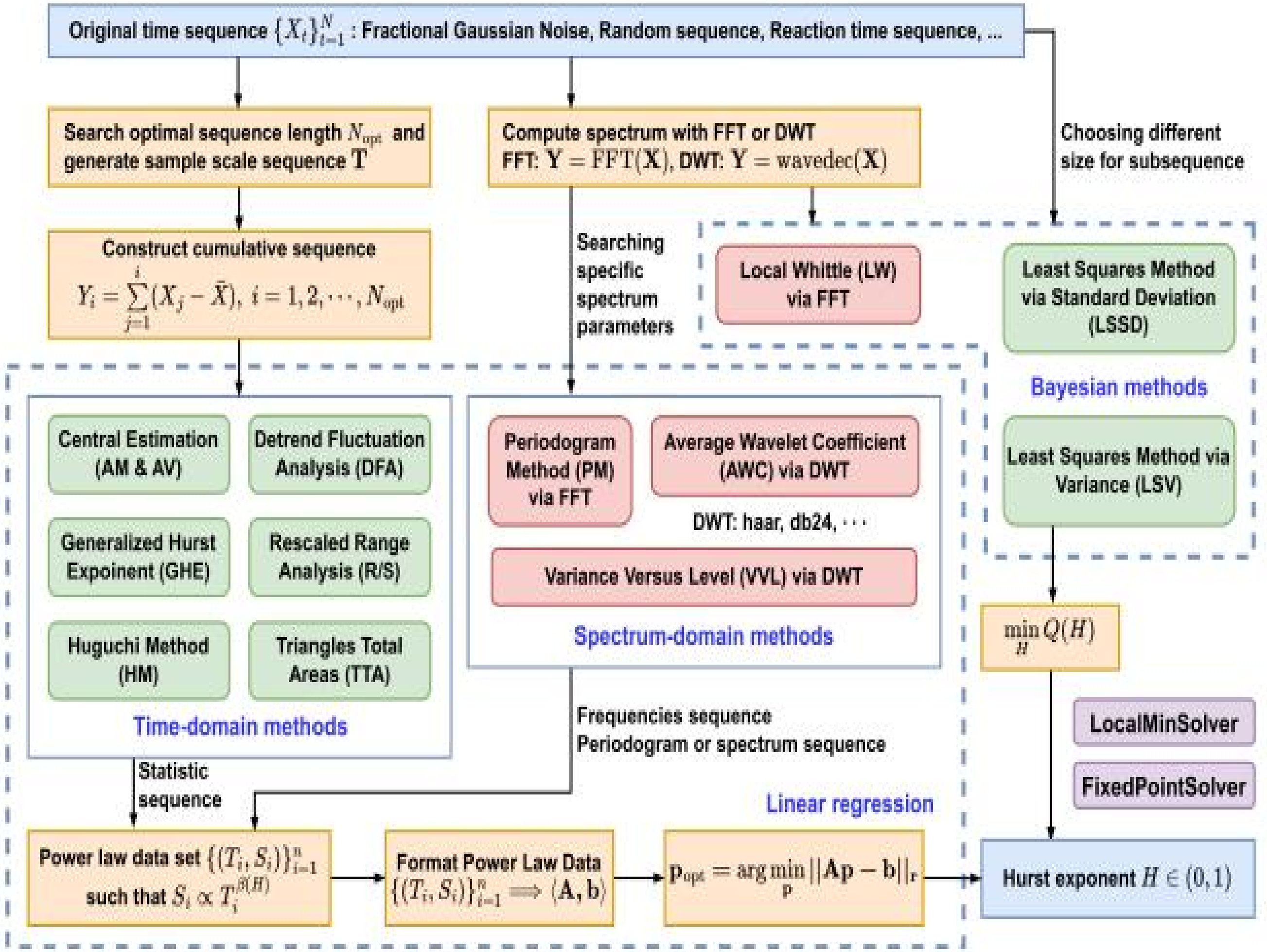
A simpler form: $VR = \frac{Var(t-k)}{kVar(t)}$

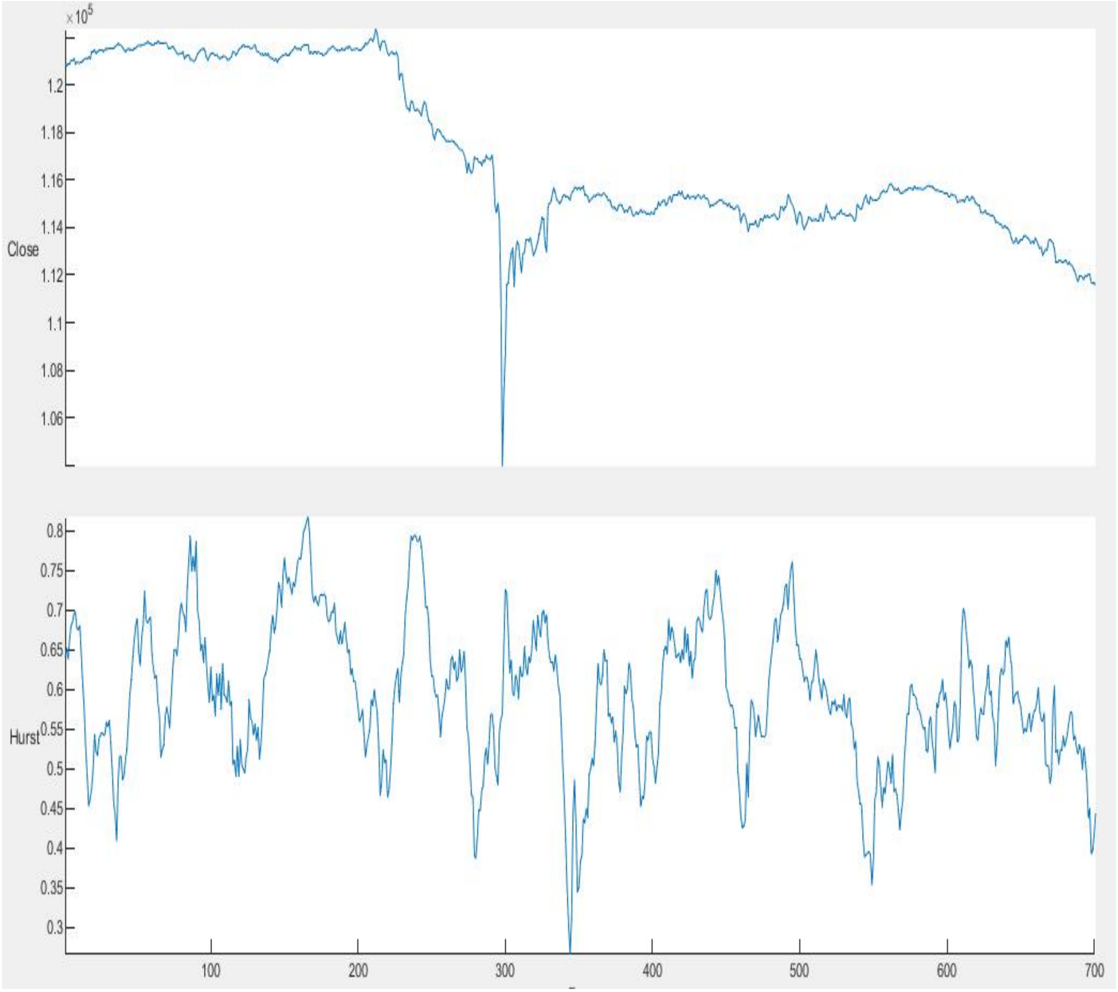


- VR=1, Random Walk
- VR>1, Momentum, ΔVar is faster than linear
- VR<1, Mean-revesion, ΔVar is slower than linear

- Lo, A. W., & MacKinlay, A. C. (1988). Stock market prices do not follow random walks: Evidence from a simple specification test. *The review of financial studies*, *1*(1), 41-66.
- https://mingze-gao.com/posts/lomackinlay1988/

2. Interdisplinary Modeling: Momentum (Example in Physics)





- *The Hurst exponent captures the autocorrelation of returns after removing the drift.*
- *If the autocorrelation function of a stationary time series decays as a power law (i.e., slower than exponential decay), then the time series exhibits long memory.*

- Hurst=0.5, Random Walk
- Hurst>0.5, Trend, long-term memory exists
- Hurst<0.5, Mean-revesion, no long-term memory

- Zhang, H. Y., Feng, Z. Q., Feng, S. Y., & Zhou, Y. (2024). Typical Algorithms for Estimating Hurst Exponent of Time Sequence: A Data Analyst's Perspective. IEEE Access.
- Kleinow, T. (2002). Testing continuous time models in financial markets. Humboldt University. Berlin.

2. Interdisplinary Modeling: Mean-Reversion (Example in Technical Analysis)

- Bollinger Band consists of a moving average line and ±2 standard deviations.
- RSI compares the magnitude of recent gains to recent losses, signaling potential reversal points.
- The performance is sensitive to the parameter selected (Time window, threshold)
- Persistence of low (high) level of RSI will bring loss to long (short) positions.
- The lower Bollinger Band is always a reliable support. A sustained price decline can pull the band downward.

2. Interdisplinary Modeling: Mean-Reversion (Example in Statistics)

Z-score:

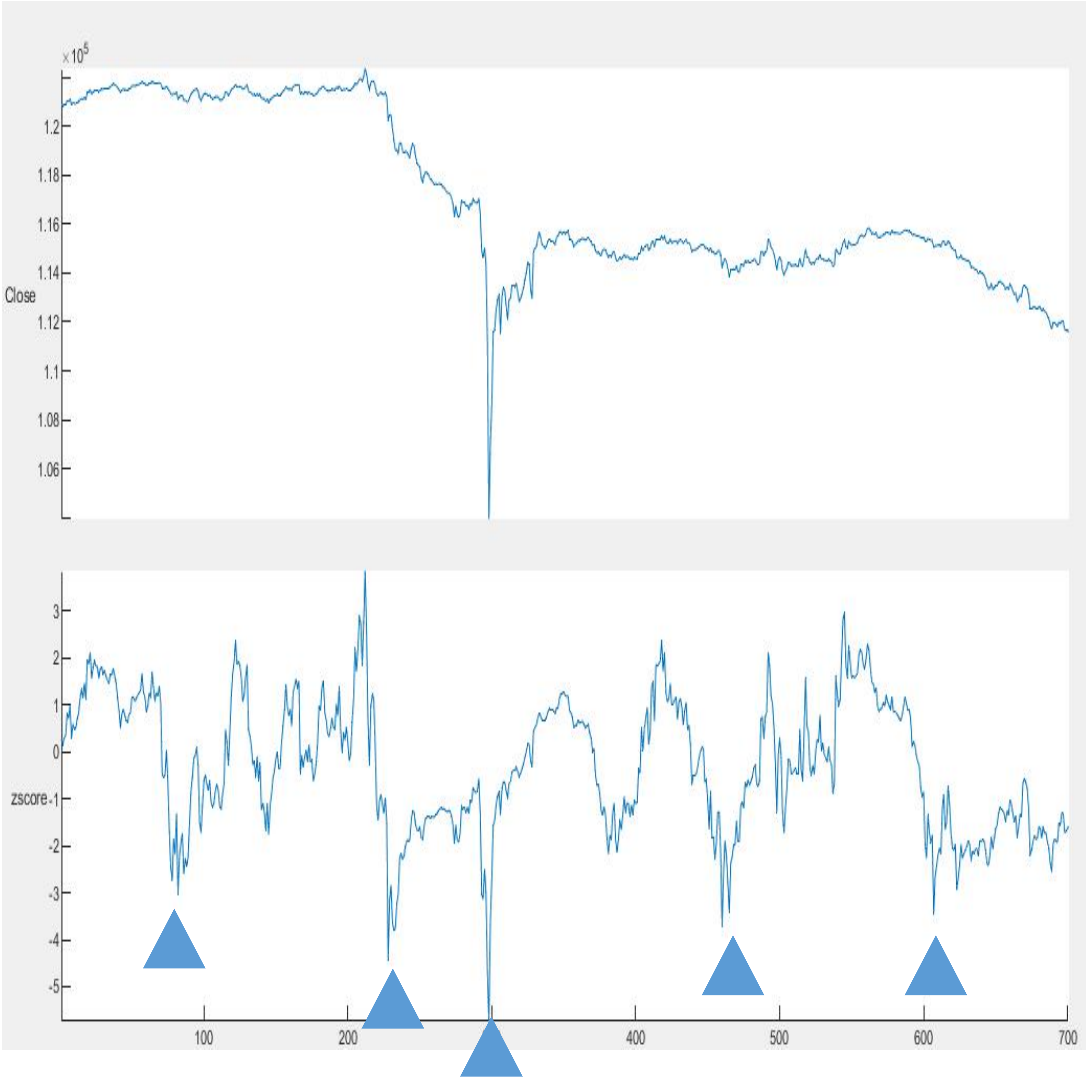
$$Zscore = \frac{(x - \mu)}{\sigma}$$

The z-score, which assumes a normal distribution, may yield unreliable results when applied to a small, or skewed dataset.

Time Series Models:

- Autoregressive integrated moving average (ARIMA)
- Exponential smoothing
- Generalized autoregressive conditional heteroscedasticity (GARCH)
- Long short-term memory (LSTM)

Stationarity is necessary for most time series models to function effectively. The Dickey-Fuller test reveals whether a dataset is stationary.



- https://www.ibm.com/think/topics/time-series-model#1190488338

2. Interdisplinary Modeling: Mean-Reversion (Example in Information Theory)

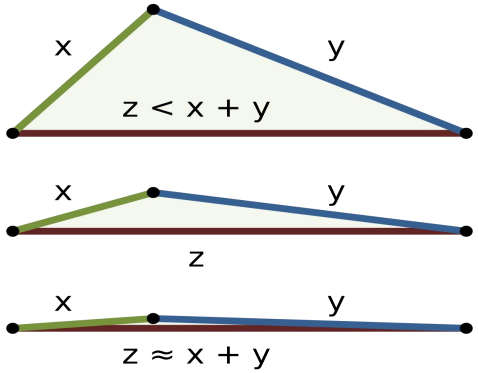
Signal-to-Noise Ratio:

$$SNR = \frac{Signal}{Noise}$$

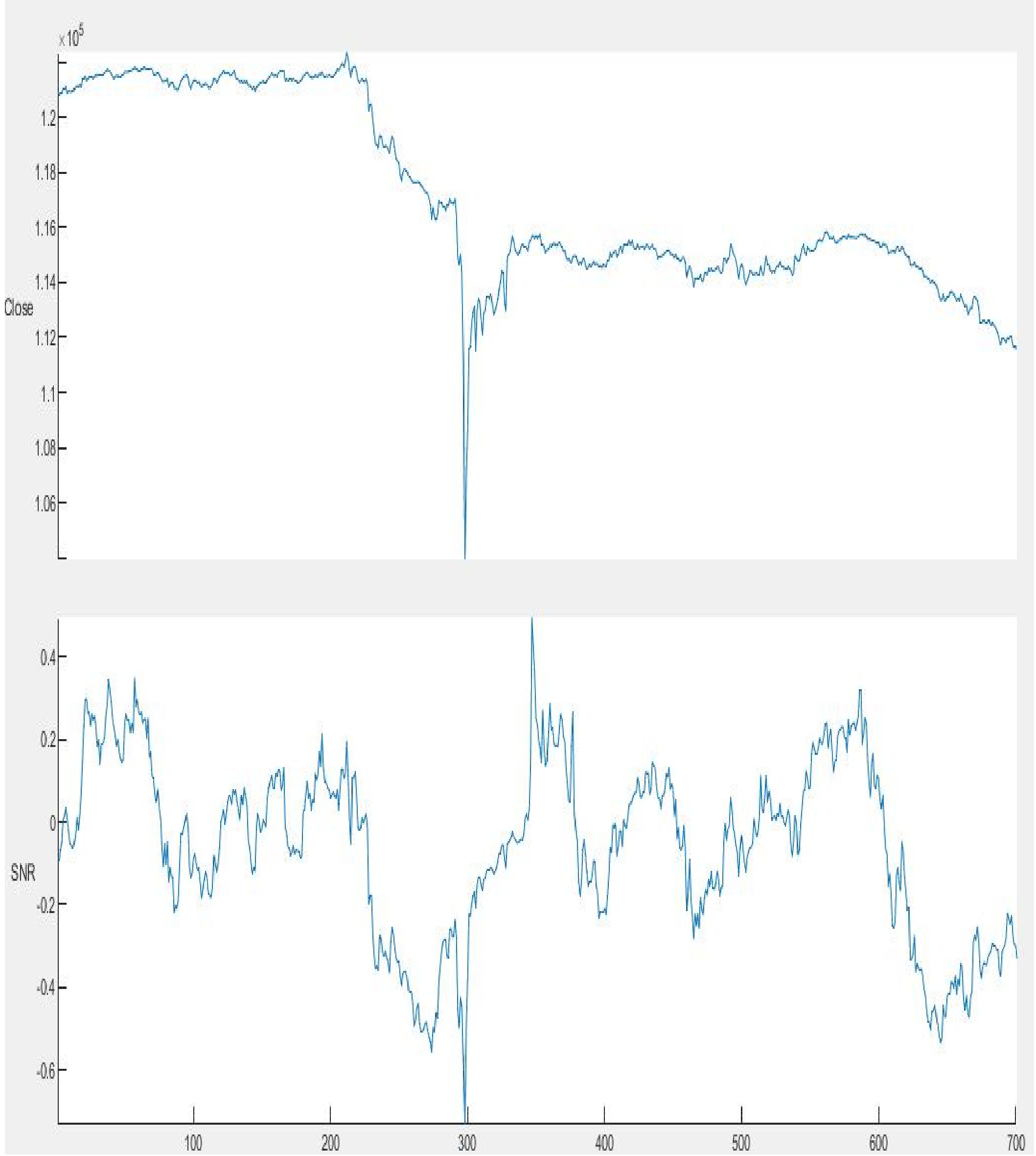
I derived my own SNR formula from triangle inequality in Euclidean geometry:

$$||x + y|| \leq ||x|| + ||y||$$

$||x + y||$ is the straight-line distance, and $||x|| + ||y||$ is the path distance.

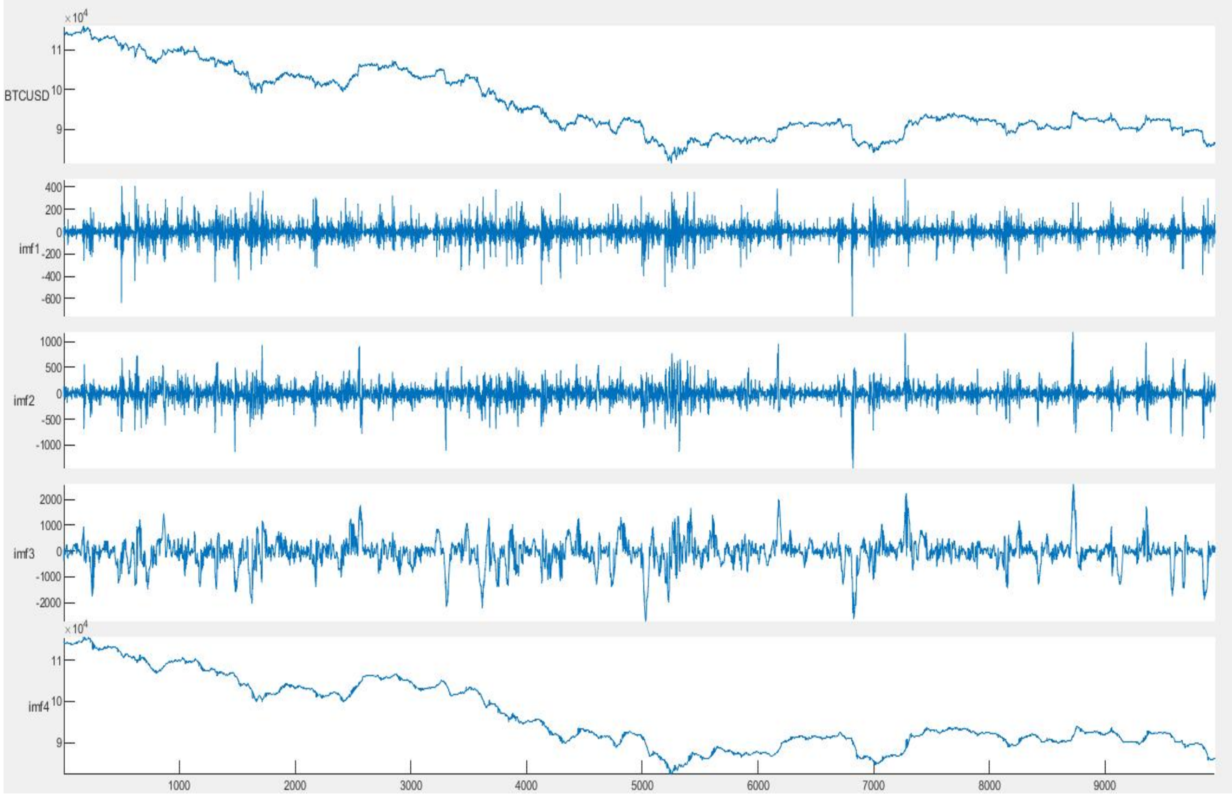


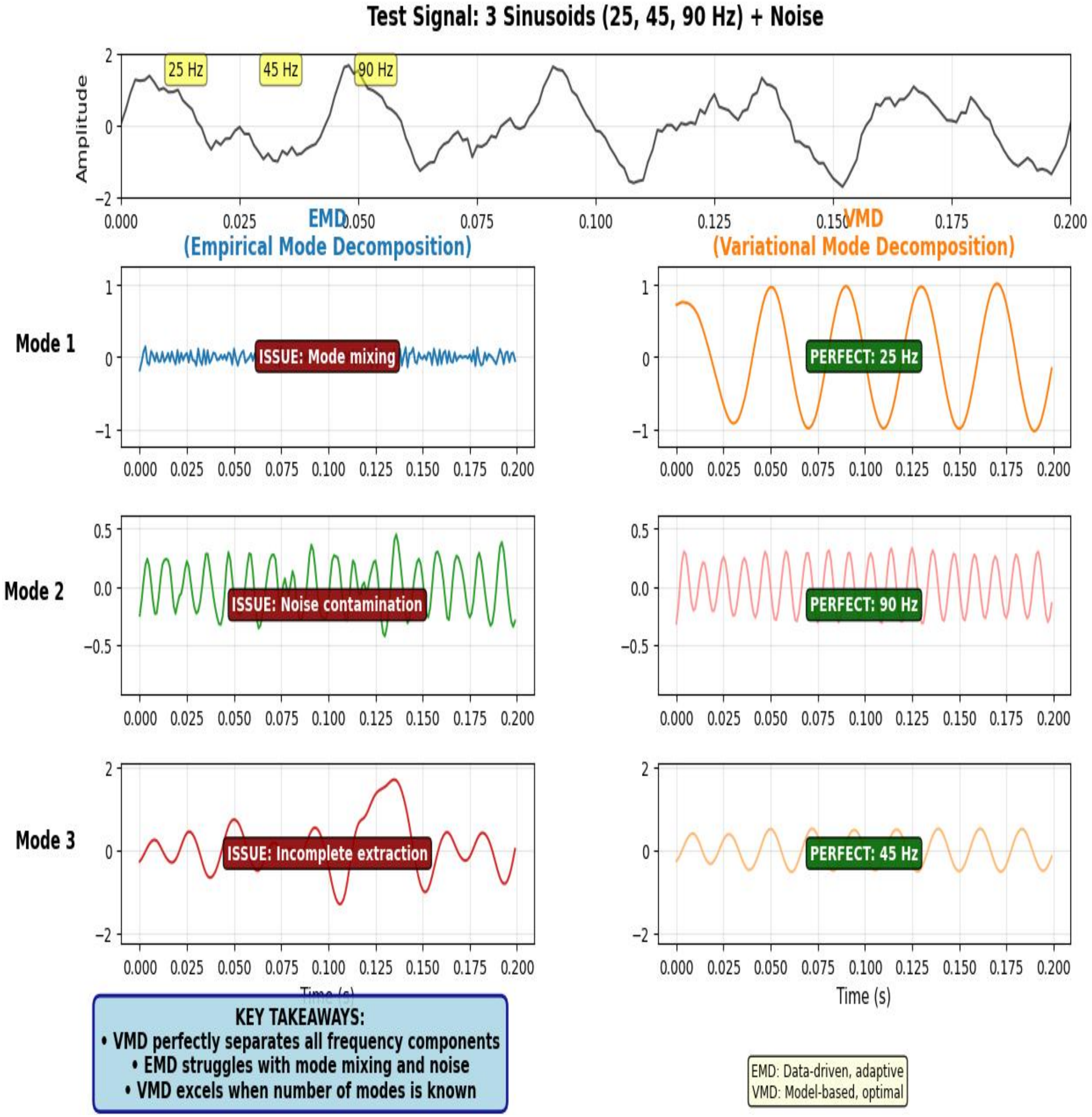
source: wiki



2. Interdisplinary Modeling: Mean-Reversion (Example in Information Theory)

Variantional Mode Decomposition (VMD):
Variational Mode Decomposition (VMD) is a non-recursive, bandwidth-penalized signal decomposition framework for extracting a prescribed number of spectrally compact intrinsic mode functions (IMFs) from **nonstationary data**.





source: https://www.kaggle.com/code/zeneilambekar/vmd-introduction

2. Interdisplinary Modeling: Mean-Reversion (Example in Physics)

In Elastic Region:
By Hooke's Law,

$$\mathrm{F} = \mathrm{k}\Delta\mathrm{L}$$

where k is the spring constant, a measure of the spring's stiffness; F is the restoring force opposes deformation

In Finance, assume

$$\Delta\mathrm{L} = \mathrm{P} - \overline{P}$$

$$\mathrm{k} = \frac{1}{\sigma}$$

then

$$\mathrm{F} = \frac{\mathrm{P} - \overline{P}}{\sigma}$$

where P , $\overline{P}$ is the current price and average price, σ is the price's volatility.

Then, F is equivalent to Z-score, and Bollinger Band. However, they may have different underlying assumptions and logics.

- If $\mathrm{k} = 1, \Delta\mathrm{L} = \mathrm{r}(0) - \mathrm{r}(\mathrm{t})$,

$F = \mathrm{r}(0) - \mathrm{r}(\mathrm{t})$ measures the momentum.

- If $\mathrm{k} = 1/\mathrm{t}, \Delta\mathrm{L} = \mathrm{P}(0) - \mathrm{P}(\mathrm{t})$,

$F = \frac{\mathrm{P}(0)-\mathrm{P}(\mathrm{t})}{t}$ is the slope.

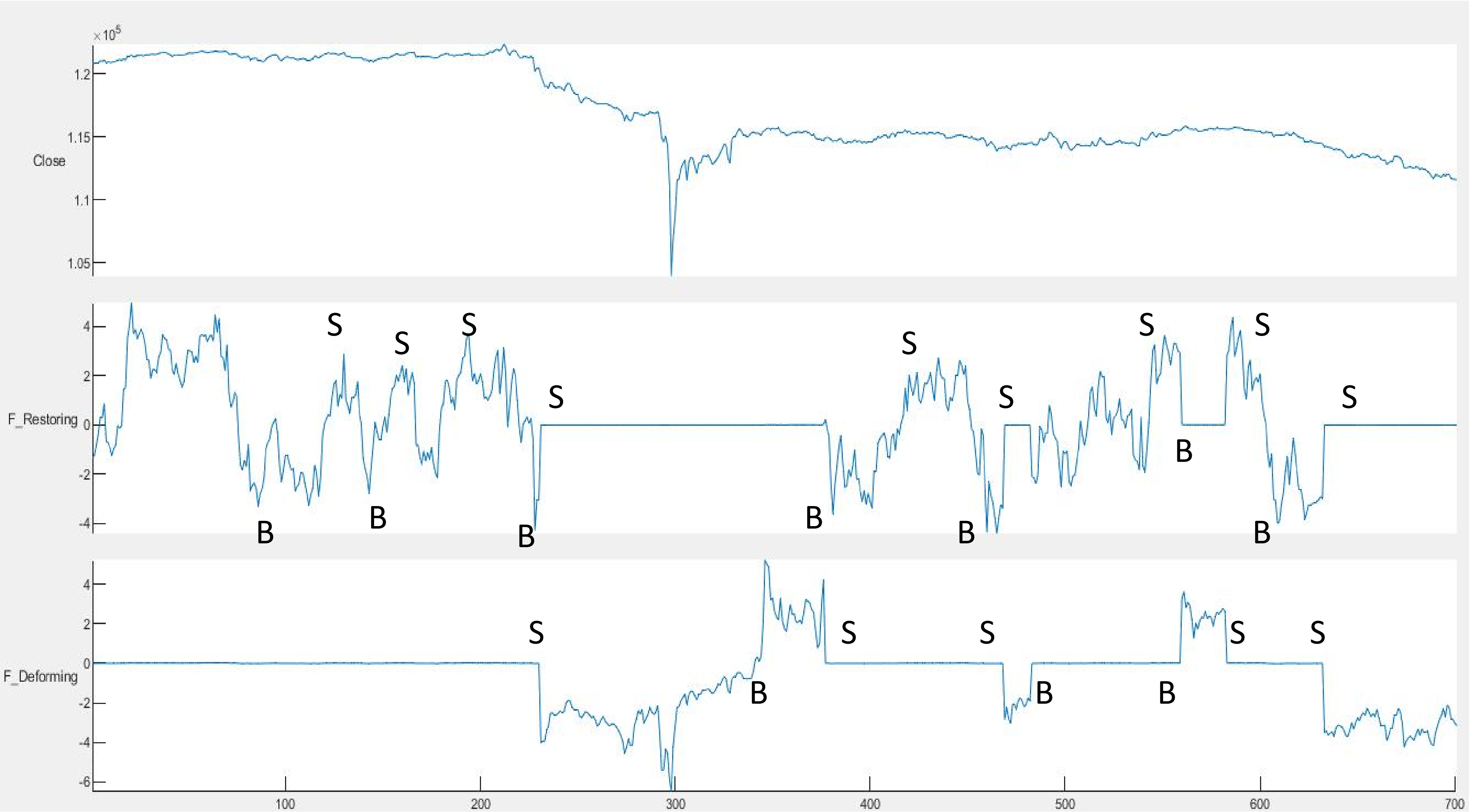
- If $\mathrm{k} = \begin{cases}\mathrm{k1}, \mathrm{x} \le \mathrm{Threshold} \\ \mathrm{k2}, \mathrm{x} > \mathrm{Threshold}\end{cases}, \Delta\mathrm{L} = \mathrm{P}(0) - \mathrm{P}(\mathrm{t})$,

It's a simplified way modeling fracture of a spring.

If defining k2<<k1, the model describes the restoring force decreases significantly when $\mathrm{x} > \mathrm{Threshold}$, meaning there's a strong trend.

If defining k1<<k2, the model describes the restoring force increases significantly when $\mathrm{x} > \mathrm{Threshold}$, such assumption helps in trading spreads between products of a similar underlying.

2. Interdisplinary Modeling: Mean-Reversion (Example in Physics)



- The signals of mean-reversion and momentum can be sperated with segamented springs.

2. Interdisplinary Modeling: Mean-Reversion (Dimension of a physical quantity)

In Elastic Region:
By Hooke's Law,

$$F = k\Delta L$$

where k is the spring constant, a measure of the spring's stiffness; F is the restoring force opposes deformation.

N=N/m*m
N=N

- If $\Delta L = P - \overline{P},\ k = \frac{1}{\sigma}$

$\Delta L = P(\$) - \overline{P}$ ($), Distance $->$ $

$$k = \frac{1\ (N)}{\sigma\ (\$)} \ -> N/\$$$

then

$$F = \frac{P - \overline{P}}{\sigma} \ -> \ N$$

- If $k = 1, \Delta L = r(0) - r(t)$, Distance$->$ %

$F = r(0) - r(t)$ measures the momentum.

$$\Delta L -> \ \%$$
$$k -> N/\%$$
$$F -> N$$

Physical laws must be dimensionally consistent.
Mapping financial data to physical variables is the most challenging step for modeling quant strategies in Physics.

e.g. In Physics,

$$F(N) = m(kg)\ a\ (\frac{meter}{second\text{^}2})$$

In our example,

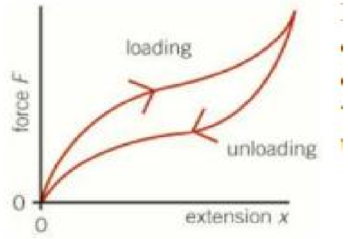
$$F(N) = m(?)\ a\ (\frac{\$}{second\text{^}2})$$
$$m -> N * second\text{^}2/\$$$

It is hard to understand or explain what is the "mass" of a stock.

2. Interdisplinary Modeling: Mean-Reversion (Example in Physics)

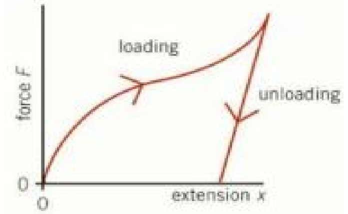
Potential Improvements:

Rubber

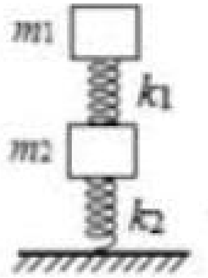


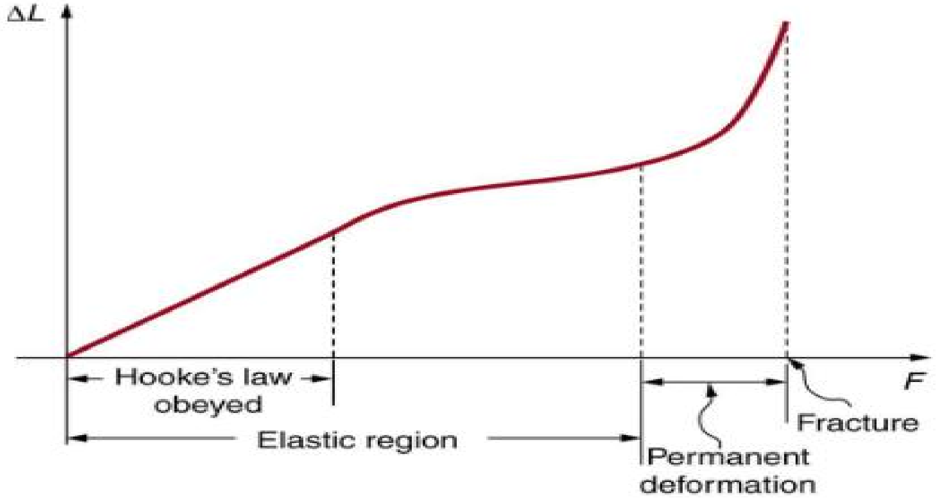
Parrallel, Horizontal

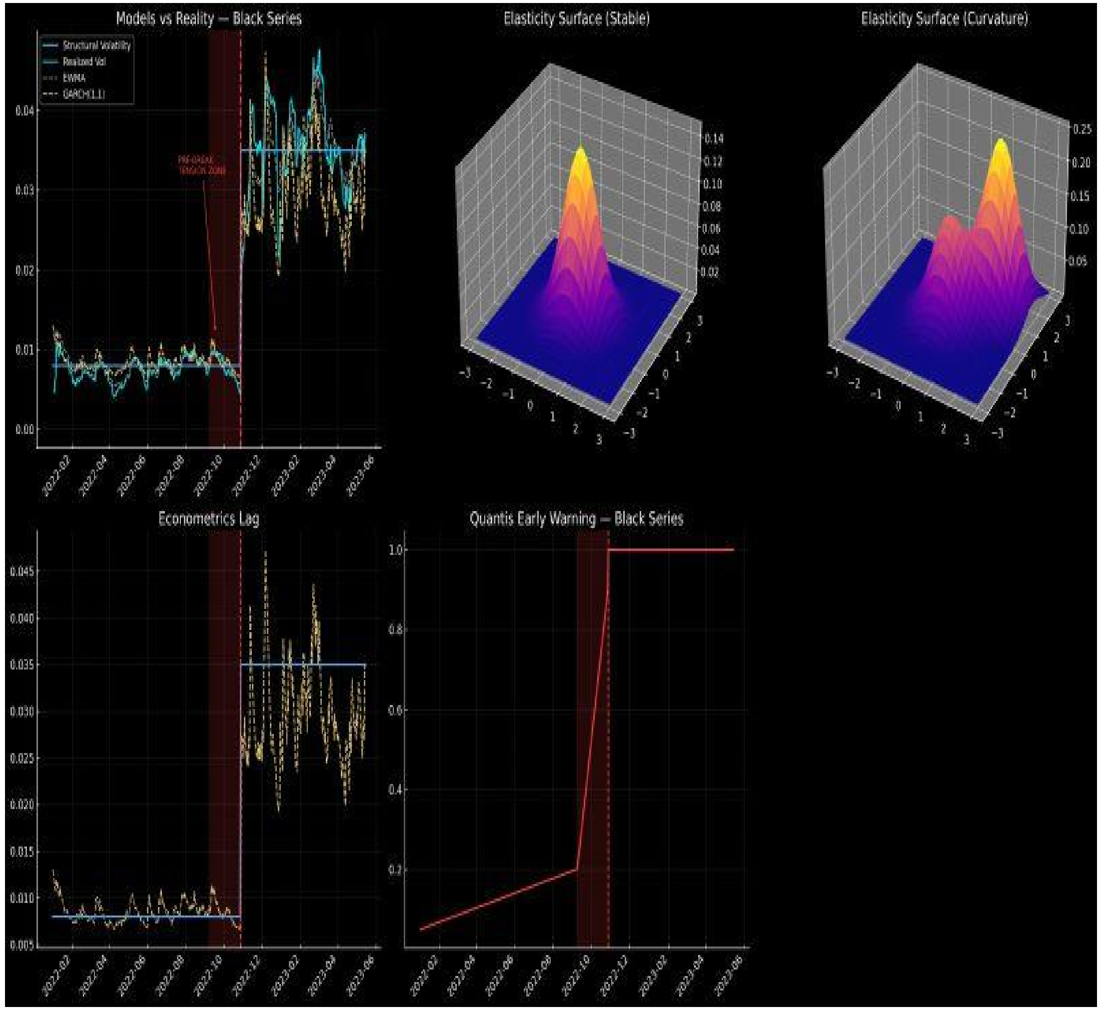
Plastic



Serial, Vertical







source:

- https://pmt.physicsandmathstutor.com/download/Physics/A-level/Notes/OCR-A/3-Forces-and-Motion/Detailed/3.4.%20Materials.pdf
- x-Douglas College Physics 1107 Fall 2019 Custom Textbook

source: Alessio Fratini, Quantis Capital

Assignment 3: Quant Modeling

Lecture 6

1. (30) Extend your trading idea into a trading strategy. Fill Table 1 (Core Logic Design of Trading Strategies) and Table 2 (Design of Algorithmic Models) showed in Lecture 5.
2. (30) Illustrate why your quant model is a good choice serving the strategy.
3. (40) Create clear and specific trading rules based on your quant model outcomes. (When to buy, when to sell, what's the optimized position, etc.)

Appendix

Empirical Test (Back Test, Forward Test, Paper Trading, Live Trading)	
Benchmark	
Performance Assessment	Sharpe Ratio, Calmar Ratio, Maximum Drawdown, Expected Return (Annuallized, Compounding/Non-compounding), Trading Frequency, Average Holding Period, Winning Probability, etc.
Back Test 1	The back-testing period 1 should contain both bull and bear market conditions.
Back Test 2	The back-testing period 2 should be the nearest 3-6 months to now.
Back Test Scenario 1	Test the strategy performance during typical risk events such as the financial crisis in 2008, trade war, etc.
Back Test Scenario 2	
Paper Trading (Forward Test)	Paper trading is a critical step in strategy development. However, it excludes real-world frictions and operational risks inherent to live markets, such as slippage, market impact, latency, partial order fills, and technical disruptions (e.g., internet disconnections or data feed failures).
Live Trading	
Capacity Test	

Table 3. Empirical Test

Appendix

Evaluation of Empirical Results	
Refine the Strategy	● Does the strategy’s return, risk, capacity, and trading frequency align with expectations? ● What caused the drawdowns, and are there methods to mitigate them? ● Would earlier or later entry/exit timing improve performance? ● Can risk be reduced or trading frequency increased while maintaining target returns? ● Can the strategy be extended to other instruments or markets to expand opportunities?
Applicable Market Conditions	Observe the best performing market conditions of the strategy. (e.g., high volatility, uptrend with low volume, etc.)
Risk/Position Management	Position control, stoploss/stopgain, hedging, risk redflags (liquidity risk, systemic risk, etc.)
Execution Risk Assessment	Assess the impact of order fill rate, execution latency, slippage, market impact, leverage, and capital gains tax on strategy performance.

Table 4. Evaluation of Empirical Results

Thank You

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